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Shopping cart coupling system for motorized scooters

Abstract

A shopping cart coupling system for motorized scooters including a base hooking assembly with a pulling assembly and a hooking assembly. Base hooking assembly includes a first eye bolt, a second eye bolt attached to a rear portion of a motorized scooter. Pulling assembly includes a first chain and a second chain attached to the first and second eye bolts by means of carabiner hooks. First and second chains are attached in a cross orientation to a front portion of a shopping cart by means of swivel hooks. The swivel hooks can be disengaged from the shopping cart to be attached to the first and second eyebolt for convenient transportation.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to shopping cart coupling systems and, more particularly, to a shopping cart coupling system for motorized scooters that includes a set of chains with attachment fasteners disposed on each end of the chains to releasably couple a shopping cart to tow it.

2. Description of the Related Art

(2) Several designs for shopping cart coupling systems have been designed in the past. None of them, however, include connected metal links with a set of fasteners attached on the edges.

(3) Applicant believes that a related reference corresponds to U.S. Pat. No. 11,383,565 issued for tow bar or scooter. Applicant believes that another related reference corresponds to U.S. Pat. No. 9,150,064 issued for scooter and cart connection device. None of these references, however, teach of a pair of chain assemblies with attachment clips disposed on each end of both chains where the chains are used to releasably couple a shopping cart to the rear of a motorized scooter allowing the scooter to tow the cart behind it.

(4) Other documents describing the closest subject matter provide for a number of more or less complicated features that fail to solve the problem in an efficient and economical way. None of these patents suggest the novel features of the present invention.

SUMMARY OF THE INVENTION

(5) It is one of the objects of the present invention to increase a user's mobility when shopping without assistance.

(6) It is another object of this invention to provide encouragement to those who may have a disability when they need to go shopping.

(7) It is still another object of the present invention to simplify a shopping routine.

(8) It is yet another object of this invention to provide such a device that is inexpensive to implement and maintain while retaining its effectiveness.

(9) Further objects of the invention will be brought out in the following part of the specification, wherein detailed description is for the purpose of fully disclosing the invention without placing limitations thereon.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) With the above and other related objects in view, the invention consists in the details of construction and combination of parts as will be more fully understood from the following description, when read in conjunction with the accompanying drawings in which:

(2) FIG. **1** represents an operational view of an exemplary embodiment of the present invention **10**.

(3) FIG. **2** shows an isometric view of one embodiment of the first eye bolt **22**, the first carabiner hook **26**, the first chain **42**, the first metal ring **62**, and the first swivel hook **68** coupled therebetween.

(4) FIG. **3** illustrates a top view of the present invention **10**, wherein the first chain **42** and second chain **44** chains are placed in a cross style when coupled.

(5) FIG. **4** is a representation of an exploded view of the base hooking assembly **20**, the pulling assembly **40**, and the hooking assembly **60**.

DETAILED DESCRIPTION OF THE EMBODIMENTS OF THE INVENTION

(6) Referring now to the drawings, where the present invention is generally referred to with numeral **10**, it can be observed that it basically includes a base hooking assembly **20**, a pulling assembly **40**, a hooking assembly **60** and various exemplary embodiments (**100**) thereof. It should be understood there are modifications and variations of the invention that are too numerous to be listed but that all fit within the scope of the invention. Also, singular words should be read as plural and vice versa and masculine as feminine and vice versa, where appropriate, and alternative embodiments do not necessarily imply that the two are mutually exclusive.

(7) Base hooking assembly **20** includes a first eye bolt **22**, a second eye bolt **24**, a first carabiner hook **26** and a second carabiner hook **28**. In an exemplary embodiment, first eye bolt **22** may be a threaded bolt with a loop at one end well known in the art. Nonetheless, it should be considered that other fasteners with a loop or a hook at one end may be suitable for first eye bolt **22**. First eye bolt **22** may have a suitable size to be attached in a rear portion of a motorized scooter **100**, wherein the first eye bolt **22** may be secured to the motorized scooter **100** by means of the threaded portion thereof. In a suitable embodiment, second eye bolt **24** may be a threaded bolt with a loop at one end well known in the art. Nevertheless, it should be considered that other fasteners with a loop or a hook at one end may be suitable for second eyebolt **24**. Second eye bolt **24** may have a suitable size to be attached in a rear portion of a motorized scooter **100**, wherein the second eye bolt **24** may be secured to the motorized scooter **100** by means of the threaded portion. First eye bolt **22** and second eye bolt **24** are placed in opposite edges of the motorized scooter **100**, wherein First eye bolt **22** and second eye bolt **24** are placed in the same horizontal axis. In a suitable embodiment first carabiner hook **26** may be a loop with a spring-loaded gate. Nevertheless, it should be considered that other fasteners like a carabiner hook with screw, a carabiner with oval straight gate or any other variation thereof may be suitable for first carabiner hook **26**. The size of the first carabiner hook **26** may be suitable to be connected to the first eye bolt **22** by means of the loop thereof. In other embodiments, second carabiner hook **28** may be a loop with a spring-loaded gate. Nevertheless, it should be considered that other fasteners like a carabiner hook with screw, a carabiner with oval straight gate or any other variation thereof may be suitable for second carabiner hook **28**. The size of the second carabiner hook **28** may be suitable to be connected to the second eye bolt **24** by means of the loops thereof.

(8) Pulling assembly **40** includes a first chain **42** and a second chain **44**. In one embodiment, first chain **42** is a serial assembly of connected links, wherein the first chain **42** may have a suitable size to be connected to the first carabiner hook **26**. The first chain **42** may be made of a plastic material, a steel material, an aluminum material, or any other suitable material that can support a pulling stress force. In a preferred embodiment, the length of the first chain **42** may be suitable to couple a

shopping cart **102** to the motorized scooter. In a suitable embodiment, second chain **44** is a serial assembly of connected link, wherein the second chain **44** may have a suitable size to be connected to the second carabiner hook **28**. The second chain **44** may be made of a plastic material, a steel material, an aluminum material, or any other suitable material that can support a pulling stress force. In one embodiment, the length of the second chain **44** may be suitable to couple a shopping cart **102** to the motorized scooter. Best depicted in FIG. **1**.
(9) Hooking assembly **60** may include a first metal ring **62**, a second metal ring **64**, a first swivel hook **66** and a second swivel hook **68**. In an exemplary embodiment, first metal ring **62** may be a cylinder ring, a keyring or any other suitable variation thereof. First metal ring **62** may have a suitable circumference and width to be connected to the first chain **42** by means of an aperture included in the first metal ring **62**. As illustrated in FIG. **2**. In a suitable embodiment, second metal ring **62** may be a cylinder ring, a keyring or any other suitable variation thereof. Second metal ring **64** may have a suitable circumference and width to be connected to the second chain **44** by means of an aperture included in the second metal ring **62**. In a preferred embodiment, first swivel hook **66** may have a spring-loaded latch that opens when pressing down the latch and closes automatically when releasing the pressure on its latch. First swivel hook **62** may have a free loop wherein the loop may rotate and the hook of the first swivel hook **62** may maintain its position. In a suitable embodiment, first swivel hook **66** may be attached to the first metal ring **42**, wherein the loop of the first swivel hook **66** is coupled to the first metal ring **62** by means of the aperture thereof. As shown in FIG. **2**. In an exemplary embodiment, second swivel hook **68** may have a spring-loaded latch that opens when pressing down its latch and closes automatically when releasing the pressure on the latch. Second swivel hook **68** may have a free loop wherein the loop may rotate and the hook of the second swivel hook **68** may maintain its position. In a suitable embodiment, second swivel hook **68** may be attached to the second metal ring **44**, wherein the loop of the second swivel hook **68** is coupled to the second metal ring **64** by means of the aperture thereof.
(10) FIG. **3** represents the assembly between the first eye bolt **22**, the first carabiner hook **26**, the first chain **42**, the first metal ring **62** and the first swivel hook **66**, wherein the first eye bolt **22**, may be attached to a rear portion of the motorized scooter **100** and the first swivel hook **66** may be attached to a front portion of the shopping cart **102**. In a preferred embodiment, the second carabiner hook **28**, the second chain **44**, the second metal ring **64** and the second swivel hook **68** are connected such as the first elements herein described. FIG. **3** illustrates the base hoking assembly **20**, the puling assembly **40** and the hooking assembly **60** connected therebetween in a cross orientation to be attached to the shopping cart **102** allowing the motorized scooter **100** to pull the shopping cart **102** in the same orientation as thereof **102** and preventing unintended lateral movements of the shopping cart **102**.
(11) The foregoing description conveys the best understanding of the objectives and advantages of the present invention. Different embodiments may be made of the inventive concept of this invention. It is to be understood that all matter disclosed herein is to be interpreted merely as illustrative, and not in a limiting sense.

Claims

1. A shopping cart coupling system for motorized scooters, comprising: a base hooking assembly including a first eye bolt, a second eye bolt, wherein said first eye bolt and said second eye bolt are attached to a motorized scooter, said hooking assembly further includes a first carabiner hook and a second carabiner hook; a pulling assembly including a first chain and a second chain, wherein said first chain and said second chain have a symmetrical length therebetween; and a hooking assembly having a first metal ring, a second metal ring, a first swivel hook and a second swivel hook, wherein said first metal ring is attached to an edge of said chain, said second metal ring is attached

to an edge end of said second chain, said first swivel hook is attached to said first metal ring and said second hook is attached to said second metal ring.

2. The shopping cart coupling system for motorized scooters of claim 1, wherein said first eye bolt and said second eye bolt have a threaded portion and a loop at one end.

3. The shopping cart coupling system for motorized scooters of claim 1, wherein said first carabiner hook and said second carabiner hook have a spring-loaded gate that opens when pushing the gate and closes automatically.

4. The shopping cart coupling system for motorized scooters of claim 1, wherein said first metal ring and second metal ring include an aperture.

5. The shopping cart coupling system for motorized scooters of claim 4, wherein said apertures need to be tensioned to increase the area of the apertures.

6. The shopping cart coupling system for motorized scooters of claim 1, wherein said first swivel hook has a spring-loaded latch that opens when pressing down the latch and closes automatically when releasing the pressure on the latch.

7. The shopping cart coupling system for motorized scooters of claim 1, wherein said second swivel hook has a spring-loaded latch that opens when pressing down the latch and closes automatically when releasing the pressure on the latch.

8. The shopping cart coupling system for motorized scooters of claim 1, wherein said first swivel hook includes a free loop that rotates independently from the first swivel hook.

9. The shopping cart coupling system for motorized scooters of claim 1, wherein said second swivel hook includes a free loop that rotates independently from the second swivel hook.

10. A shopping cart coupling system for motorized scooters, comprising: a base hooking assembly including a first eye bolt, a second eye bolt, wherein said first eye bolt and said second eye bolt are attached to a rear portion of a motorized scooter, said hooking assembly further includes a first carabiner hook coupled to said first eye bolt and a second carabiner hook coupled to said second eye bolt; a pulling assembly including a first chain and a second chain, wherein said first chain and said second chain have a symmetrical length therebetween, said first chain is attached to said first carabiner hook and said second chain is attached to said second carabiner hook; and a hooking assembly having a first metal ring, a second metal ring, a first swivel hook and a second swivel hook, wherein said first metal ring has an aperture, said first metal ring is attached to an edge of said first chain, said second metal ring has an aperture, said second metal ring is attached to an edge of said second chain, said first swivel hook has a spring-loaded hatch, said, said first swivel hook is attached to said first metal ring and said second hook has a spring-loaded hatch, said second hook is attached to said second metal ring.

11. A shopping cart coupling system for motorized scooters, consisting of: a base hooking assembly including a first eye bolt, a second eye bolt, wherein said first eye bolt is attached to a rear portion of an edge of a motorized scooter, said second eye bolt is attached to a rear portion of an opposite edge of said motorized scooter, said first eye bolt and said second eye bolt are attached in the same horizontal axis, said hooking assembly further includes a first carabiner hook coupled to said first eye bolt and a second carabiner hook coupled to said second eye bolt; a pulling assembly including a first chain and a second chain, wherein said first chain and said second chain have a symmetrical length therebetween, said first chain is attached to said first carabiner hook, said second chain is attached to said second carabiner hook, said first chain and said second chain are made of a steel material; and a hooking assembly having a first metal ring, a second metal ring, a first swivel hook and a second swivel hook, wherein said first metal ring has an aperture, said first metal ring is attached to an opposite edge of said first chain, said second metal ring has an aperture, said second metal ring is attached to an opposite edge of said second chain, said first swivel hook has a spring-loaded hatch, said, said first swivel hook has a free loop that rotates independently, wherein said first metal ring is attached to said free loop of said first swivel hook, said second swivel hook has a free loop that rotates independently, wherein said second metal ring is attached to said free loop of

said second swivel hook, said second hook has a spring-loaded hatch, said first swivel hook and said second swivel hook are attached to a shopping cart.